

Perceiver & Perceiver IO

Implementation and Experimental Analysis

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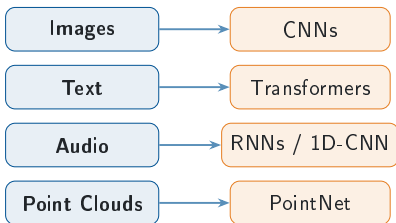
Università degli Studi di Firenze
Course B031278 — Deep Learning

Agenda

- 1 Introduction
- 2 Perceiver
- 3 Experiments: Perceiver
- 4 Perceiver IO
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- 6 Conclusion

The Problem: Modality-Specific Architectures

One modality, one architecture



Each data type is usually paired with a dedicated model and domain-specific assumptions about structure.

The scaling bottleneck

Self-attention cost

$$\mathcal{O}(M^2 \cdot d)$$

grows quadratically with input length M

Input representation	M
16×16 image patches	196
raw 224×224 pixels	50,176

Raw pixels make standard attention
impractical.

Core question: can one architecture handle arbitrary inputs without quadratic cost?

The Perceiver Solution

Core idea: learned latents re-read the input T times; expensive processing stays in latent space.

1. Read

Cross-attention

M inputs $\rightarrow N \ll M$ latents

2. Process

Latent block

transform N latent slots

3. Reuse

Shared weights

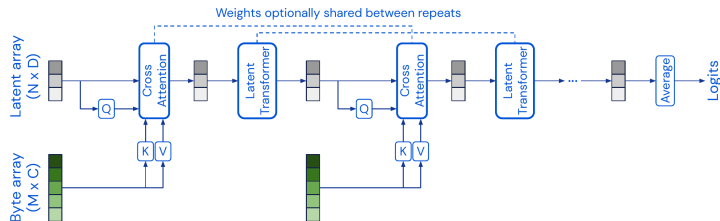
repeat the same processor

Complexity shift

Read the full input T times; repeat the expensive work in the compact latent space $N \ll M$.

Operation	Cost
Read input (per read)	$\mathcal{O}(M \cdot N)$
Process latents for L layers	$\mathcal{O}(L \cdot N^2)$
Perceiver	$\mathcal{O}(MN + LN^2)$

Perceiver Architecture



Encode and process

- $X \in \mathbb{R}^{M \times C}$ supplies keys and values.
- $Z \in \mathbb{R}^{N \times D}$ supplies queries, with $N \ll M$.
- Cross-attention maps $X \rightarrow Z$: $\mathcal{O}(MN)$.

Latent core and output

- Latent Transformer updates Z only.
- Repeated blocks may share weights.
- Original Perceiver pools Z ; Perceiver IO uses output queries.

Cross-Attention: The Key Mechanism

Formally:

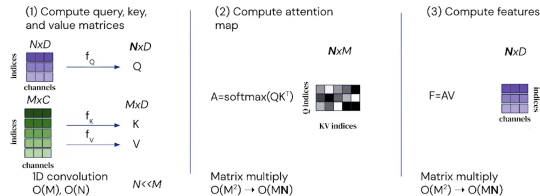
$$Q = Z W_Q \in \mathbb{R}^{N \times d_k}$$

$$K = X W_K \in \mathbb{R}^{M \times d_k}$$

$$V = X W_V \in \mathbb{R}^{M \times d_k}$$

$$\text{CrossAttn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

The **latents Z are the queries**, the input X provides keys and values $\Rightarrow$ informational *bottleneck*.

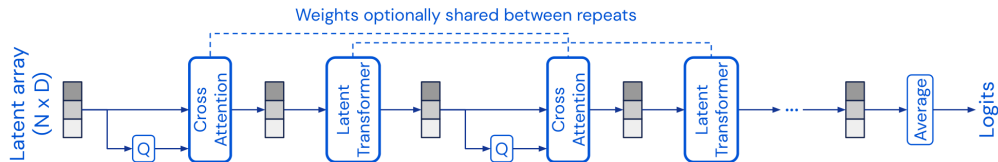


Key insight:

- Attention matrix: $N \times M$ (not $M \times M$)
- Complexity: $\mathcal{O}(N \cdot M \cdot d_k)$
- Decouples depth from input size

Latent Transformer & Weight Sharing

Once information is in the latent array, depth is spent by reusing the same processor on a compact state.



Weight sharing

- The same Latent Transformer block B_θ is reused at every repeat, with shared parameters θ .
- Unrolling depth behaves like an RNN over the latent state: $Z_{t+1} = B_\theta(Z_t)$.

Why it matters

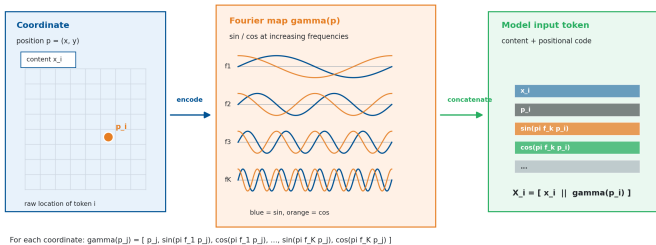
- Iterative refinement without instantiating a new processor at each step.
- Parameter count stays fixed as depth grows; computation stays in the N -slot latent space.

Fourier Positional Encoding

Perceiver receives a flat array, so location has to be part of each input token.

Fourier features turn position into a multiscale signature

The input stays a flat array: every element receives content plus its encoded coordinates.



General recipe

Use domain coordinates: images (u, v) , point clouds (x, y, z) , or text index t ; sine/cosine bands encode them at multiple scales.

Input seen by attention

The array is still flat, but each token is enriched before projection: $X_i = [x_i \parallel \gamma(p_i)]$.

Perceiver Forward Pass: Shape Checkpoints

Data flow in the original Perceiver:

- 1 Input array: each token stacks content x with a Fourier position code $\gamma(p)$,

$$X = [x \parallel \gamma(p)] \in \mathbb{R}^{M \times C}.$$

- 2 Learned latent array:

$$Z_0 \in \mathbb{R}^{N \times D}, \quad N \ll M$$

- 3 Cross-attention reads input:

$$Q = ZW_Q, \quad K = XW_K, \quad V = XW_V$$

- 4 Latent Transformer applies self-attention + FFN only on N latents.

Shape checkpoints:

Object	Shape	Meaning
X	$M \times C$	input array (content+pos)
Z_0	$N \times D$	learned latents
Q	$N \times d_k$	latent queries
K, V	$M \times d_k$	input keys/values
A	$N \times M$	attention map

Core idea

The input size M affects only the cross-attention read; depth is spent in the compact latent space.

Original Perceiver Output: Pooling Classifier

Fixed-shape output head:

$$Z_T = \text{Encoder}(X, Z_0)$$

$$z = \frac{1}{N} \sum_{i=1}^N Z_T[i]$$

$$\text{logits} = zW_{\text{cls}} + b_{\text{cls}}$$

- Mean pooling collapses all latents to one vector
- The classifier predicts one global label
- This is ideal for classification, not dense structured outputs

Where FFN and sharing fit:

- Each latent block is pre-norm attention, residual, FFN, residual
- FFN adds non-linearity after attention mixing
- Weight sharing repeats the processor over iterations

Fixed-shape output

Pooling collapses the N latents into a single vector, so this head emits one global prediction per input.

Computational Complexity Analysis

Model	Self-Attn	Cross-Attn	Total
Standard Transformer	$\mathcal{O}(M^2 d)$	—	$\mathcal{O}(M^2 d \cdot L)$
Perceiver	$\mathcal{O}(N^2 d)$	$\mathcal{O}(MNd)$	$\mathcal{O}((MN + N^2 L) d)$

Symbols:

- M : number of input elements
- N : number of latent slots, with $N \ll M$
- L : number of latent processing layers
- d : hidden channel dimension

Interpretation:

- The full input is read through cross-attention
- Repeated self-attention happens only in latent space
- If N is fixed, depth becomes independent of input length
- Concrete sizes are evaluated in the experiments

GELU Activation & LAMB Optimizer

GELU (Gaussian Error Linear Unit):

$$\text{GELU}(x) = x \cdot \Phi(x) \approx x \cdot \sigma(1.702 x)$$

- Smooth activation used in Transformer architectures
- Allows small negative gradient flow

LAMB (Layer-wise Adaptive Moments):

$$r_t^{(l)} = \frac{\min(\|w_t^{(l)}\|, 10)}{\|\hat{u}_t^{(l)}\|}$$

Per-layer trust ratio rescales updates and stabilizes large-batch training.

Why LAMB for Perceiver?

- Paper (p. 6): easier to optimise than SGD, the CNN default
- The paper never compares LAMB with Adam
- Made for large batches; here batch 64 plus clipping

Optimizer	Perceiver paper	Layer-wise LR
SGD	Harder to optimise	Global
LAMB	Easier (p. 6)	Per-layer trust

Implementation

Aspect	Paper implementation	Project code
Core model	Cross-attention into learned latents, latent self-attention, FFN, weight sharing	Same modules; weight sharing is exposed as an ablation switch
Model scale	Large runs: latents up to 2048×512 (IO flow); IO text 201M, IO++ 425M params	Smaller runs: CIFAR-10 96×384 , ModelNet40/text 128×512
Data setting	ImageNet, ModelNet40, large language pretraining corpora	CIFAR-10, ModelNet40, WikiText-103 byte-level MLM, GLUE fine-tuning
Training budget	Distributed accelerators, batches 512–8192	Single-GPU budget, batches 32–64, LAMB stable only with clipping at 1.0
Practical additions	Minimal regularization in the reference setup	Validation split carved from train, fixed seed, attention-map extraction, declarative run registry

The implementation keeps the architectural mechanism intact; besides scale, data and compute, some design choices differ too (next slide).

Implementation Differences from the Paper

Choice	Original paper	This project
Input tokenization	Raw pixels, no patching ($M = 50,176$)	Raw pixels, no patching ($M = 1,024$ on 32×32)
Latent array $N \times D$	512×1024	96×384 (CIFAR-10), 128×512 (ModelNet40)
Cross-attend iterations T	8	4 (CIFAR-10), 2 (ModelNet40) — and T is itself an ablation axis
Positional encoding	Fourier, $K = 64$ bands/axis (ImageNet: 258 dims; ModelNet40 too); text: learned	CIFAR-10 identical : 258 dims + 3 RGB = 261. ModelNet40 and text: $K = 6$ ($3 + 39 = 42$ per point, $257 + 13 = 270$ per byte)
Augmentation, masking	ModelNet40: scale jittered <i>per point</i> (shift: no gain). MLM: 15% of <i>whole words</i> masked	Scale and shift of the <i>whole</i> cloud, undone by the normalisation. MLM: 15% of <i>single bytes</i> — an easier task
Weight sharing	On by default (shared across T)	Exposed as an ablation switch (on/off)
Model selection	—	Validation carved from train (45k/5k/10k); test touched once. ModelNet40 has no validation split: epoch picked on test
Parameters, MLP width	44.9M on ImageNet; MLP $1 \times$ (hidden = D)	10.18M (CIFAR-10), 23.29M (ModelNet40), 18.87M (IO text); MLP $4 \times$

Kept identical to the paper

Core mechanism preserved: cross-attention bottleneck $\rightarrow$ weight-shared latent transformer $\rightarrow$ mean-pooling (Perceiver) / output-query decoder (IO); GELU MLP (but $4 \times$ wide, the paper uses $1 \times$), pre-norm LayerNorm, Fourier frequencies linearly spaced to Nyquist.

CIFAR-10: Experimental Setup

Goal: validate the Perceiver on raw pixels, with no convolutional assumption anywhere.

Hyperparameter	Value
Latent array ($N \times D$)	96×384
Cross-attn stages (T)	4, interleaved
Transformer blocks (L)	4 (shared)
Heads (enc. cross / self)	1 / 8
Fourier bands / max freq	64 / 16
Dropout	0
Optimizer	LAMB
Learning rate	0.004
Scheduler	MultiStep 84/102/114
Batch size	64
Epochs	120, <i>no early stop</i>
Precision	bfloat16
Total parameters	10 175 362

Protocol — the part that makes the numbers usable

Train split: **45 000** images. Validation: **5 000**, carved out of the training set. Test: the official **10 000**, touched *once*, at the very end.

The reported checkpoint is the one with the best *validation* accuracy. Selecting the epoch on the same set you report on is **selection leakage** and inflates every number.

Data pipeline

- 32×32 RGB image $\rightarrow$ **1 024 pixels** as an unordered set
- Each pixel: 3 RGB + 2D Fourier PE = **261 dims** — the same per-pixel width as the paper
- No patching, no convolution, no learned tokenizer

The Noise Band: How Much Does a Number Move on Its Own?

Three runs differing in seed and validation split land 2.78 points apart. Read every other comparison against that.

Three runs, differing in seed and validation split:

Run	Seed	Test acc.
e01_baseline	42	71.63%
e31_baseline_seed1	1	68.85%
e32_baseline_seed2	2	70.97%
Spread (max – min)		2.78pp

The rule for every slide that follows

Any difference smaller than **2.78 percentage points** cannot be told from noise. Seed and split alone can produce it.

Why this slide exists

Two extra runs cost a few GPU-hours and change the status of all the others: without a measured band, every comparison is an opinion.

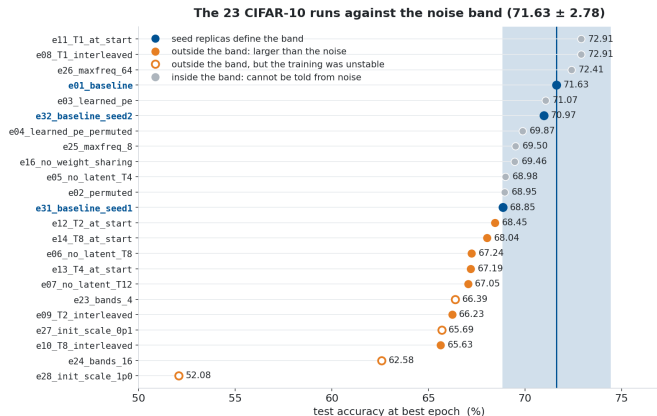
It also sets the honest ceiling of this study. On CIFAR-10 with 1 024 pixels and one GPU, most of the paper's ablations — measured on ImageNet with 50 176 pixels on TPU pods — simply do not have room to show up.

The band depends on the metric: 2.78pp on test accuracy, 2.94pp on best validation, 3.24pp on final validation. Borderline verdicts can flip (e13 on validation).

e01 is the *luckiest* of the three seeds. Measured from their mean (70.48%) instead, e12 and e14 fall back inside: 10 runs clear the band, not 12.

CIFAR-10: What Survives the Noise Band

Ranked against the band: twelve of the twenty-one ablations clear it, the other nine are inconclusive.



21 ablations, 12 outside

10 against the three-seed mean

- **Outside** the band: no PE, init scale, T with depth, arrangement at $T = 4$, no latent transformer, bands (collapsed runs)
- **Inside**: permutation, learned PE, weight sharing, max frequency

The chart plots 23 runs: e29_no_pe at 32.36% is off scale. Source: each run's results.json, test accuracy at the validation-selected epoch.

CIFAR-10: Permutation Invariance — the Headline Claim

Shuffling the pixels costs nothing measurable. Removing the encoding costs 39.27 points. That contrast is the claim.

Comparison	Positional encoding	Natural → permuted	Δ
e01 → e02	Fourier	71.63% → 68.95%	−2.68
e03 → e04	learned	71.07% → 69.87%	−1.20
e01 → e29	none at all	71.63% → 32.36%	−39.27
<i>Paper (ImageNet):</i> Fourier 78.0 → 78.0 learned 70.9 → 70.9			

What we measured

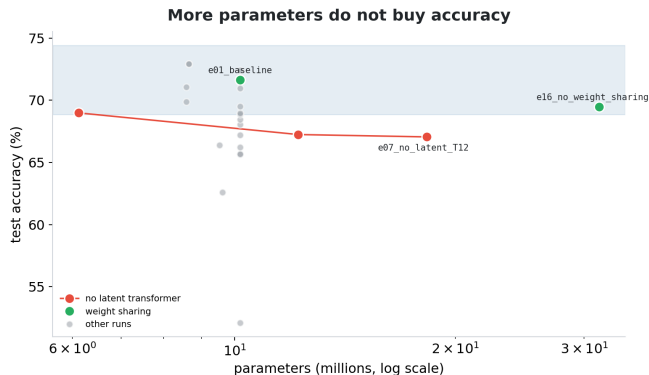
Both drops fall **inside** the 2.78pp band: shuffling the pixels causes no measurable damage. Expected by construction: the PE is attached *before* shuffling and attention ignores order, so e02 computes the same function; −2.68 is noise.

The control that makes it a claim

Shuffling the pixels costs nothing measurable; **removing the encoding costs 39.27pp**. The same architecture that ignores the order collapses without the coordinates. Position enters through the **encoding**, never through the grid — and that contrast, not either number alone, is the evidence.

CIFAR-10: Capacity Does Not Buy Accuracy

Three times the parameters do not buy a point: capacity is not what limits this model at CIFAR-10 scale.



Unsharing latent blocks: no gain

- e01 71.63% with 10.2M params
- e16 69.46% with **31.5M** params
- $3.1\times$ the weights, **-2.17pp** — inside the band

Where you put capacity matters

- Remove the latent transformer, add cross-attends instead:
- 6.1M $\rightarrow$ 68.98%
- 12.2M $\rightarrow$ 67.24%
- 18.3M $\rightarrow$ 67.05%
- **All below e01; no trend**

CIFAR-10: How Many Cross-Attends, and Where

T	interleaved	all at start	Δ (arrangement)
1	72.91%*	72.91%*	0.00
2	66.23%	68.45%	+2.22
4	71.63% (baseline)	67.19%	−4.44
8	65.63%	68.04%	+2.41

The only arrangement effect outside the band

At $T = 4$, interleaving beats front-loading by **4.44pp** — outside the band on test (3.29pp from the mean), inside on validation from the mean. At $T = 2$ and $T = 8$ the sign flips (inside the band): the paper's rationale stays a hint.

Across rows, reads and depth change together: each read reuses the same 4 shared latent blocks (depth $4T$: 4 at $T = 1$, 32 at $T = 8$), while the paper keeps 48 fixed. Within a row the comparison is clean. *Inside the band, at depth 4 instead of 16.

A reproducibility check, for free

At $T = 1$ the two arrangements produce the *same computational graph*, and indeed give bit-identical results: 72.91%, epoch 90, 8 654 662 params. Fixed seed $\Rightarrow$ reproducible training.

CIFAR-10: Latent Init and Fourier Sweeps

Sweep	Value	Test acc.
Latent init σ	0.02	71.63%
	0.1	65.69%
	1.0	52.08%
Fourier bands	4	66.39%
	16	62.58%
	64	71.63%
Max frequency	8	69.50%
	16	71.63%
	64	72.41%

Latent init: monotone here, unlike the paper's Fig. 6

σ from 0.02 to 1.0 costs **19.55pp**. At $\sigma = 1.0$ the best epoch is **9 out of 120**: training then collapses (10.6% at epoch 71); in the paper $\sigma = 1.0$ costs ~ 1 point.

Two honest caveats

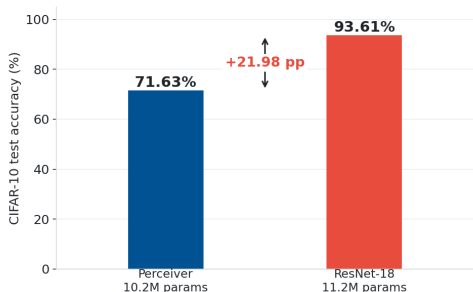
Bands is non-monotone (4 beats 16), and both low points are collapsed trainings (e23: 10.1% at epoch 84, e24: 13.1% at epoch 59) — they measure the collapse, not the bands.

Max frequency sits entirely inside the noise band. The whole axis is inconclusive, and saying so is part of the result.

CIFAR-10: the Reference That Is Not a Perceiver

Neither Perceiver paper benchmarks CIFAR-10, so we trained the reference ourselves: a ResNet-18, same split and epochs, its own SGD recipe.

Same data, split and epochs; each with its own recipe (LAMB vs SGD)



The uncomfortable comparison comes first

A **ResNet-18** on the same split, the same 120 epochs and 11.2M parameters against 10.2M reaches **93.61%**, where e01_baseline reaches **71.63%**: **-21.98pp**.

Neither Perceiver paper benchmarks CIFAR-10, so without this run there is nothing to read the number against.

What the gap actually measures

On ImageNet the paper is only *on par* with ResNet-50 (78.0 vs 77.6; the 73.5 is ResNet-50 on Fourier features). Here the gap also reflects a different recipe (SGD vs LAMB), not only the architecture.

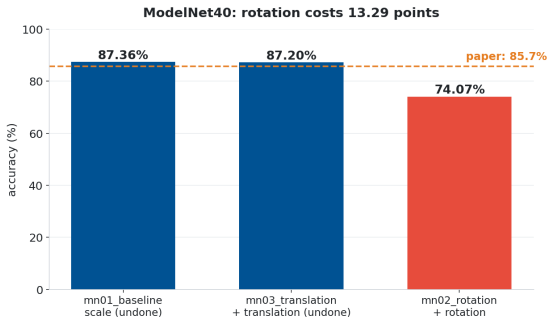
The Perceiver's claim was never accuracy — it is **having no prior on the domain**. Here prior and recipe together are worth 22 points.

ModelNet40: Same Architecture, 3D Point Clouds

The same architecture, fed 2,048 points in 3D instead of pixels. Sizes, bands and recipe change too.

Setting	Value
Latent array	128×512
Cross-attn / blocks	2 / 6
Points per model	2 048
Input per point	42 dims (3 + 39)
Fourier bands / max f	6 / 1120 (paper: 64)
Optimizer / lr	LAMB / 0.001
Batch (paper: 512)	32
Epochs	120
Parameters	23 288 928

Metric note: ModelNet40 has no separate validation split — the standard test set (2 468 objects) is the validation set, so the figure is `val_accuracy` at the best epoch. The last epoch, with no selection at all, gives **86.43%**.



Above the paper

87.36% vs the paper's **85.7%**, with a batch $16\times$ smaller — and still above at the last epoch. But the setup differs (6 bands, lr steps) and there is no noise band: the point is generality, not the margin.

ModelNet40: Rotation Costs 13 Points — and Why

Rotation costs 13.29 points — the positional encoding again. Scale and translation never reach the network.

Augmentation	Accuracy	Δ
scale only*	87.36%	—
+ translation*	87.20%	−0.16
+ rotation	74.07%	−13.29

* Two of the three runs see the same data

The normalisation re-centres each cloud and rescales it to unit radius, so a scale or a shift of the *whole* cloud is undone before the model sees it — the paper jitters *each point* instead. `mn01` and `mn03` are therefore a free replica: 0.16pp is run-to-run noise (same seed 42).

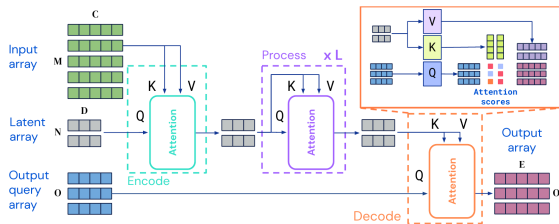
The Perceiver has no rotational inductive bias

3D Fourier PE encodes **absolute** position. Rotating an object changes every coordinate — and therefore every encoding — while the label stays the same. The network must learn from scratch an invariance the architecture does not grant, and that this task does not even need: ModelNet40 objects are probably stored upright (not verified).

Here the axis is arbitrary in 3D and the rotation is applied in training only: the test objects stay upright. The paper also found that rotation did not help.

Takeaway: learning an invariance the test set does not need costs accuracy.

Perceiver IO: Structured Outputs



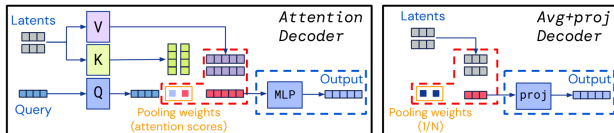
Extension over Perceiver (Jaegle et al., ICLR 2022):

- **Encode:** input $\rightarrow$ latent (same cross-attn encoder)
- **Process:** latent self-attention ($\times L$)
- **Decode:** output queries read latents

Why it matters:

- Original Perceiver: one pooled global output
- IO: task-specific output shape
- Same latent bottleneck, more general decoder

Perceiver IO Decoder with Output Queries



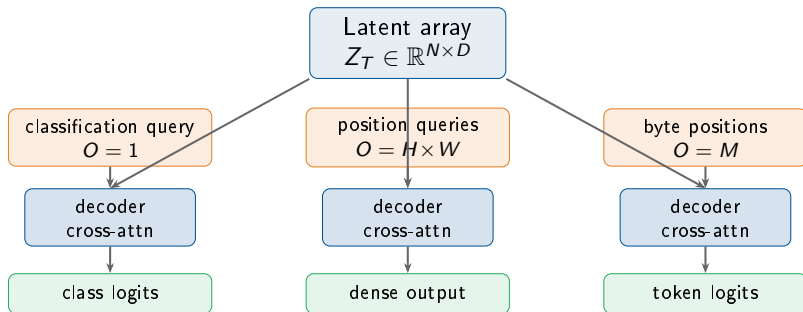
Decoder cross-attention (single layer):

$$Q_{\text{dec}} = \text{OutputQuery } W_Q, \quad K_{\text{dec}} = \text{Latent } W_K, \quad V_{\text{dec}} = \text{Latent } W_V$$

$$\text{Output} = \text{softmax}\left(\frac{Q_{\text{dec}} K_{\text{dec}}^T}{\sqrt{d_k}}\right) V_{\text{dec}} \in \mathbb{R}^{O \times D}$$

- A task head then projects each of the O rows to the output channels E (logits, pixels, bytes)
- The number of output queries O is **task-dependent** and independent of N and M
- A single decoder cross-attention layer suffices (no self-attention among outputs)

Output Query Design per Task



Classification

One learned query produces one global prediction.

Dense outputs

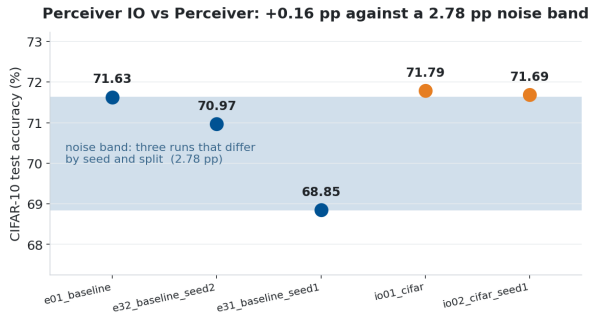
Output shape is controlled by the query array.

Masked LM

One query per byte; the loss counts only the masked ones.

Perceiver IO on CIFAR-10: No Measurable Difference

On a single label out of ten, query decoder and mean pooling land inside the noise band — with decoder and recipe both changed.



What changed, and what did not

Same encoder as e01_baseline; the mean-pooling head is replaced by a **decoder with one output query**, and the recipe changes too: $\text{lr } 10^{-3}$ cosine (the IO paper, App. A.1: $2 \cdot 10^{-3}$ flat, then cosine).

io01_cifar **71.79%** vs e01 **71.63%**: +0.16pp, against a seed-and-split spread of **2.78pp**. The replica io02 lands at 71.69%.

What the result says, and what it does not

No measurable difference is the result, not a disappointment. The decoder earns its keep when the output is **structured** — one query per byte in the MLM, one per task in the multitask run. For one label out of ten, the decoder's edge in the paper (+0.4 with Fourier, IO Tab. 7) is below our band.

WikiText-103: Byte-Level Masked Language Model

Pre-train a language encoder with no tokenizer at all: the input is the raw UTF-8 byte stream.

Setting	Value
Latent array	128×512
Cross-attn / blocks	1 / 4
Heads (enc. cross / self)	1 / 8
Sequence length	512 bytes
Vocabulary	$256 + 1$ mask
Masking	15% of single bytes
Output queries	512 (8 heads)
Optimizer / lr	LAMB / 0.001
Epochs / batch	10 / 32
Parameters	18 872 740
Masked-byte accuracy	86.68%
Lookup of the two neighbours	42.96%
Always a space (most frequent)	19.07%
Uniform chance (1/256)	0.39%

Where the decoder finally earns its keep

Nearly nine masked bytes out of ten, reading raw UTF-8 with **no tokenizer at all** — **44 points above** a lookup table of the two neighbouring bytes. So the model must use more than the two neighbours.

Why bytes are affordable here

Cross-attention costs $O(M \cdot N)$ — *linear* in sequence length — instead of $O(M^2)$. Going from 512 tokens to 2 048 bytes costs BERT 16 $\times$ in attention, and Perceiver IO only 4 $\times$.

Read this number carefully

We mask *single bytes*, the paper *whole words*: a masked byte usually keeps its neighbours, which is why the lookup already reaches 42.96% (`baseline_mlm.py`). Nor is it comparable to masked-*token* accuracy — the comparison with BERT lives on GLUE.

GLUE: Eight Tasks, One Encoder, No Tokenizer

The pre-trained encoder, a new query and output layer — and two of the eight numbers sit at or near the majority class.

Run	Task	Budget	Result
io_glue_cola	acceptability	30	69.13
io_glue_sst2	sentiment	10	80.73
io_glue_mrpc	paraphrase	30	70.59
io_glue_stsb	similarity (reg.)	30	60.82
io_glue_qqp	duplicate Q	3	84.66
io_glue_mnli	NLI (3-way)	3	65.80
io_glue_qnli	QA entailment	10	80.65
io_glue_rte	entailment	30	56.68
Mean over the eight			71.13
<i>Paper, Tab. 1 (byte-level)</i>			<i>81.0</i>

STS-B is Pearson $\times 100$, as in the paper. Grey: at or just above the majority class. Fine-tuning starts from the MLM checkpoint; the epoch budget follows task size (393 k examples against 2.5k).

Two numbers sit at or near the majority class

CoLA scores 721 correct out of 1 043 — *exactly* the majority-class count. **MRPC** beats its majority by 9 examples out of 408. CoLA is a constant predictor, MRPC barely more.

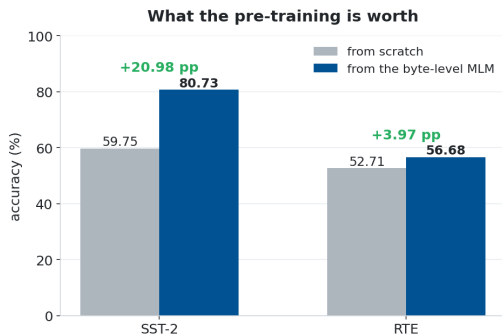
The mean of 71.13 therefore rests on **six** tasks, not eight. In the paper's metric (Matthews) CoLA scores **0**, and our mean drops to **62.49**.

The gap that does not close

62.49 (Matthews) against the paper's 81.0. The scale is not comparable: **18.9M** parameters pre-trained on WikiText-103 for 10 epochs, against **201M** on Wikipedia + C4 for 500 000 steps.

GLUE: What the Pre-training Is Actually Worth

Two tasks fine-tuned twice — once from the MLM checkpoint, once from random weights. Everything else identical.



RTE from scratch = majority class (146/277): RTE is not learned well either way

Why two controls and not eight

A fine-tuned 80% means nothing on its own — maybe the task is solvable at 80% from random weights. The two scratch runs are chosen **far apart** in data size: **SST-2** has 67 000 training examples, **RTE** has 2 500. Everything else is identical.

And the prediction holds

SST-2: 59.75% → 80.73%, **+20.98pp**. RTE: 52.71% → 56.68%, **+3.97pp**.

Transfer is worth five times more on SST-2 than on RTE. RTE is not learned well either way: from scratch it sits at its majority class (52.71%).

GLUE Multitask: Same Sign as the Paper, No Noise Band

One model with eight output queries against eight separate fine-tunings — the comparison from Table 2 of the IO paper.

One model with eight output queries vs eight models: 74.05 vs 71.13 (one run each)

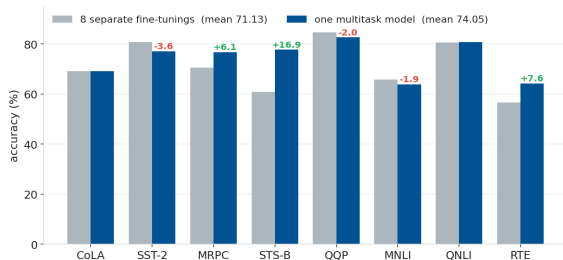


Table 2 of the IO paper

One model with **one output query per task**, against eight separate fine-tunings. Paper: 81.8 vs 81.0, +0.8. Here: 74.05 vs 71.13, +2.92.

The absolute level stays ~16 points below with CoLA in Matthews (65.41). The **sign** matches Tab. 2, but with one run per task and no noise band it is a hint, not a measurement.

Where the gain sits

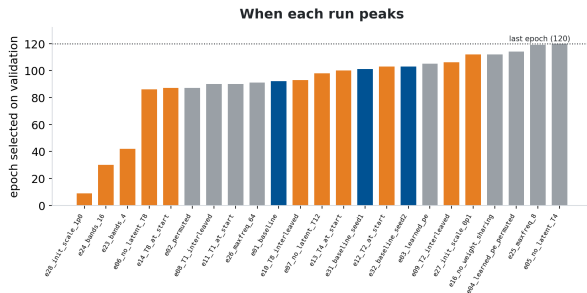
All on the small tasks — STS-B +16.9, RTE +7.6, MRPC +6.1 — which see only 1.35 passes over their own data. SST-2, QQP and MNLI lose 1.9 to 3.6 points, QNLI gains 0.2; CoLA stays at the majority class in both.

Not a fair fight — against the multitask model

The multitask model selects **one** epoch for all eight tasks; the separate models select **eight**. It still scores higher (one run, no noise band); a ninth task costs one query and one linear head.

Convergence: When Each Run Peaks

There is no early stopping: every run trains the full schedule and reports the checkpoint that was best on validation.



Why this matters

The distribution of best epochs is itself a diagnostic: a run that peaks early and then decays is unstable, whatever its best number says.

Four unstable runs

Validation accuracy, best epoch → final epoch:

e23: 67.74% @42 → **22.54%**

e24: 62.66% @30 → **55.46%**

e28: 53.36% @9 → 46.32%

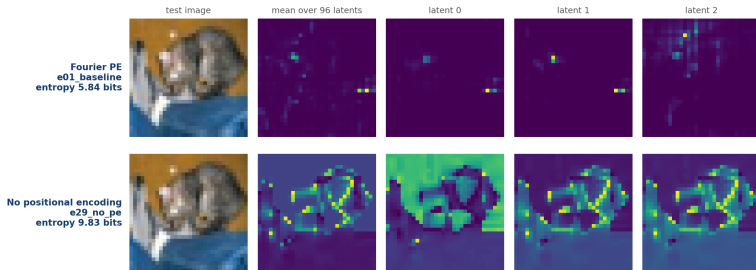
e27: 63.00% @30 → **25.32%** @52, recovers

Their reported numbers come from the validation-selected checkpoint — the correct methodology — but the instability has to be declared, not hidden.

e05_no_latent_T4 peaks at epoch 120, but its validation was flat (69.0–69.4%) from epoch 103.

Attention Maps: Where the Latents Look

With coordinates the latents pick locations; without, they can only pick colours.



Fourier PE (e01, 71.63%)

Each latent attends to a few pixels at specific locations: mean entropy **5.84 bits** over 1 024 pixels (uniform: 10 bits). One image; latents overlap.

No PE (e29, 32.36%)

Same-colour pixels are indistinguishable: attention follows colour boundaries and spreads over the image — **9.83 bits**, 98% of uniform.

Comprehensive Results Summary

Forty-two runs, all completed. Every number below traces to a log, a results.json or the papers.

Modality	Model	Best Acc.	Reference	Experiments
Images (CIFAR-10)	Perceiver	72.91% (in band)	93.61% [†]	24 runs
Images (CIFAR-10)	Perceiver IO	71.79%	71.63% (Perceiver)	2
Point Clouds (ModelNet40)	Perceiver	87.36%	85.7%	3
Text (WikiText-103)	Perceiver IO	86.68% MLM	42.96% lookup	1
NLU (GLUE, 8 tasks)	Perceiver IO	71.13 acc. / 62.49 MCC	81.0 (paper)	11
Images (CIFAR-10)	ResNet-18	93.61%	—	1

[†] A ResNet-18 trained here: same data, split and epochs, own SGD recipe. Neither Perceiver paper benchmarks CIFAR-10, so the reference had to be measured rather than quoted.

What is measured

- ModelNet40 above the paper: +1.66pp (+0.73 last epoch), other setup, no band
- Byte-level MLM: 86.68%, 44pp above a neighbour lookup
- One multitask model vs eight separate: 74.05 vs 71.13, no noise band
- Noise band **2.78pp**, from 3 replicas (seed and split)
- 12 of 21 CIFAR ablations clear the band (10 against the seed mean)

Scope of the campaign

- Declarative registry of **42 runs**, **42** completed
- 3 modalities: 2D images, 3D point clouds, raw bytes
- One command per run: `experiments.py -run <id>`
- Every figure above traces to a log, a results.json or the papers

Gap vs Original Papers

On the modality with the weakest prior we land above the paper (other setup). On the one with the strongest, a convolutional net of the same size beats us.

ModelNet40 (Point Clouds):

Source	Acc.
Original Paper	85.7%
This implementation	87.36%
Gap	+1.66pp

We are above the paper.

With a batch $16\times$ smaller (32 vs 512). But the setup differs (6 bands, lr steps) and there is no noise band. The epoch is picked on the test set; the last one, 86.43%, is still above.

What the two columns say together

On the modality with a **weak** domain prior — 3D point clouds — we are above the paper. On the one with a **strong** prior, a convolutional net of the same size wins by 22 points. On ImageNet the paper is on par with ResNet-50 (78.0 vs 77.6); we cannot test scale here: what we replicate is **generality**, not accuracy.

CIFAR-10 (Images):

Source	Acc.
ResNet-18, own SGD recipe	93.61%
e01_baseline	71.63%
Gap	-21.98pp

The reference had to be measured.

Neither Perceiver paper benchmarks CIFAR-10, so a convolutional net was trained here (same data, split and epochs; own SGD recipe) rather than quoted.

Hardware Limitations & Design Choices

Every limitation below is one GPU instead of a TPU pod. The mechanism is the paper's; the budget is not.

Parameter	Original Papers (DeepMind)	Project Setup
Hardware	TPUs (Perceiver: not stated; IO: 64–256)	1 NVIDIA RTX 3080
Batch size	512–8192	32–64
Latents (N)	512 (ImageNet), 2048 (IO flow)	96–128
Model size	44.9M (Perceiver), 201M (IO text)	10.2–23.3M params
Pre-training corpus	Wikipedia + C4, 201M-param IO	WikiText-103, 18.9M-param IO

Consequences:

- LAMB at batch 32–64, not the 512–2048 of the papers
- 96–128 latents instead of 512
- One seed per ablation, three for the baseline
- No search beyond the registered ablations

Mitigation strategies:

- Weight sharing: 10.2M parameters, not 31.5M
- bf16 autocast + gradient clipping (LAMB at lr 0.004)
- A measured noise band, not more runs per cell
- Validation carved from train; CIFAR-10 test touched once

Possible Improvements

Architecture:

- Increase latent count ($N = 256+$) with gradient checkpointing
- Add multi-scale cross-attention (coarse $\rightarrow$ fine)
- Explore relative positional encodings (RoPE, ALiBi)
- Deeper decoder for structured outputs

Training:

- Larger batch sizes with gradient accumulation
- More seeds per ablation, for a better estimate of the noise band
- Longer training schedules
- Advanced augmentation (CutMix, MixUp)

Data & Tasks:

- ImageNet pre-training for vision
- Longer byte-level pre-training: more epochs, larger corpus
- Multi-modal joint training
- Audio modality experiments

Analysis:

- Layer-wise attention probing
- Latent space clustering visualization
- Systematic hyperparameter search
- Comparison with efficient Transformers (Linformer, Performer)

Empirical Summary

Everything measured, split by whether it survived the noise band.

Large effects (noise band measured on CIFAR-10 only):

- **Positional encoding** is the whole thing: removing it costs **39.27pp** (71.63→32.36%)
- **Generality**: one architecture, 87.36% on 3D point clouds — above the paper — and 86.68% on raw bytes
- **Pre-training transfers**: SST-2 59.75→80.73%, and 5× less on the small task
- **Latent init scale**: 0.02→71.63%, 1.0→52.08%, via unstable training (paper: −1.2)
- **Interleaving** beats front-loading by 4.44pp, at $T = 4$ only, at the band's limit
- **Rotation augmentation** costs 13.29pp on ModelNet40

Inside the band — “no measurable damage”:

- Permutation, with *both* positional encodings: expected by construction
- Unsharing latent blocks ($3.1\times$ the parameters): no measurable gain
- Perceiver IO vs Perceiver on images: +0.16pp
- ModelNet40 translation: −0.16pp — undone by the normalisation, a free replica

Measured, and not in our favour:

- A ResNet-18 with its own SGD recipe: **93.61%** vs 71.63%
- GLUE mean 62.49 in Matthews (71.13 in accuracy) against the paper's 81.0
- CoLA is a majority-class predictor; MRPC beats its majority by 9 of 408
- No configuration beats the baseline outside the band — every conclusive effect is **negative**

Central methodological finding

Measuring the **noise band** — three replicas differing in seed and split — cost two runs and changed the status of all the others. Without it, 9 of the 21 ablations (11 against the seed mean) would have been reported as effects, though indistinguishable from noise.

Conclusions

What the architecture gives, and what it still costs at this scale.

Architecture properties:

- Single architecture for images, point clouds, and text
- Linear in input size (by construction, not measured)
- Unsharing latent blocks ($3.1\times$ the parameters): no measurable gain
- Zero CNN/RNN domain-specific assumptions
- Output queries: one more task costs **one query** and one linear head, not another model

Remaining limitations:

- At this scale prior and recipe together are worth **22 points** (93.61 vs 71.63)
- Entirely dependent on the positional encoding (-39.27 pp without it)
- GLUE ~ 18.5 below the paper in Matthews (62.5 vs 81.0), 1/10 of the parameters
- Latent init scale destabilises our training (paper: 0.1 diverges, 1.0 costs ~ 1 pt)
- Half the image ablations are indistinguishable from noise

Closing statement

A **single general-purpose architecture** does apply to images, point clouds and raw bytes — and on the modality with the weakest domain prior it lands above the paper, with another setup. Its advantage was never accuracy: it is **having no prior on the domain**. At CIFAR-10 scale prior and recipe together are worth 22 points; on ImageNet the paper is on par with ResNet-50, and nothing here contradicts it.

Thank you!

Questions?

Francesco Gigli and Corso Vignoli

Code and reproduction instructions shared with the course